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Boosters, hedges and the negotiation of academic knowledge

Introduction

The expression of doubt and certainty is central to the rhetorical and interactive character of academic writing. Its importance lies in the fact that academics gain acceptance for their research claims by balancing conviction with caution, either investing statements with the confidence of reliable knowledge, or with tentativeness to reflect uncertainty or appropriate social interactions. Following Holmes (1983, 1984, 1990), I will refer to this as hedging and boosting. However, while a range of studies have demonstrated the pragmatic importance of hedging as a resource for expressing uncertainty, scepticism and deference in academic contexts (Hyland, 1996a & b, 1998a; Salager-Mayer, 1994; Skelton, 1988; 1997), we still know little about how it functions, or is typically realised, in specific academic domains. Even less is known about the role of firm assertion, a potentially face-threatening strategy which seems to contradict the need to maintain a harmonious relationship with the reader.

In this paper I draw on interviews with academic writers and an analysis of research articles from eight academic disciplines to offer a preliminary characterisation of the importance, distribution and functions of hedges and boosters in a range of fields. The paper is organised in terms of both the range of features employed in the disciplinary corpora and the major functions they perform in them. I begin by presenting the quantitative results, then go on to examine the distribution of features in terms of the disciplinary conventions of hard and soft knowledge domains. In the longest section I consider the role of authorial involvement in rhetorical choices, examining how the strategic management of commitment influences such issues as writer presence and participant relationships. Before I discuss the study, however, I would like to briefly outline the general discoursal role of hedges and boosters.

What are hedges and boosters?

Hedges and boosters are communicative strategies for increasing or reducing the force of statements. Their importance in academic discourse lies in their contribution to an appropriate rhetorical and

interactive tenor, conveying both epistemic and affective meanings. That is, they not only carry the writer's degree of confidence in the truth of a proposition, but also an attitude to the audience.

Boosters such as *clearly, obviously* and *of course*, allow writers to express conviction and assert a proposition with confidence, representing a strong claim about a state of affairs. Affectively they also mark involvement and solidarity with an audience, stressing shared information, group membership, and direct engagement with readers. In the following typical case, the writer employs a series of boosters to underline the conviction he wishes to attach to his argument:

> (1)This brings us into conflict with Currie's account, for static images surely cannot trigger our capacity to recognize movement. If that were so, we would see the image as itself moving. With a few interesting exceptions we obviously do not see a static image as moving. Suppose, then, that we say that static images only depict instants. This too creates problems, for it suggests that we have a recognitional capacity for instants, and this seems highly dubious. (Phil)

Hedges, like *possible, might* and *perhaps*, on the other hand, represent a weakening of a claim through an explicit qualification of the writer's commitment. This may be to show doubt and indicate that information is presented as opinion rather than accredited fact, or it may be to convey deference, humility, and respect for colleagues views (Myers, 1989; Hyland, 1996b, 1998a). Again, these are also often found in clusters, although here they act to reinforce the uncertainty of the writer's propositions, or at least the degree of certainty that it may be prudent to attribute to them:

> (2) Our results suggest that rapid freeze and thaw rates during artificial experiments in the laboratory may cause artifactual formation of embolism. Such experiments may not quantitatively represent the amount of embolism that is formed during winter freezing in nature. In the chaparral at least, low temperature episodes usually result in gradual freeze-thaw events. (Bio)

These examples illustrate common distributional patterns where either hedges or boosters tend to cluster together in 'modally harmonic' combinations (Lyons, 1977:807) to express a kind of epistemic concord running through a series of clauses or sentences. However, hedges and boosters can also be found together in stretches of discourse where writers seek to create different rhetorical effects. It is clear in the following extracts, for example, what propositions the writers consider to be established knowledge, and what they regard as more contentious, the combination of hedges and boosters contrasting the epistemic validity of different parts of an argument to more effectively present their

claims. In (3) this acts to strengthen the specific research findings contrasting two models while limiting the writers commitment to the more general claim; in (4) the writer begins with what can be safely accepted and then moves on to highlight novelty and draw more tentative conclusions from experimental data:

> (3) Our results suggested that Moffitt's developmental theory specifying two higher-order latent factors may explain the underlying structure of antisocial behavior across the early life course, from age 5 to age 18. In a test of a general theory against a developmental theory using parent reports, the two-factor model was clearly supported over the single-factor model. Additionally, two conceptual replications using self reports and teacher reports demonstrated the utility of the two-factor model. (Mkt)

> (4) Although it is clear that some group II introns are spliced efficiently under physiological conditions only if aided by trans-acting factors, it remains plausible that others may actually self-splice in vivo. Our results indicate that the splicing of nearly every pre-mRNA intron in the maize chloroplast genome requires either chloroplast ribosomes or crs2 function. The splicing of this intron may require nuclear gene products not yet identified in our genetic screens. Alternatively, this intron may self-splice in vivo. (Bio)

A better idea of the rhetorical work accomplished by these devices can be appreciated by referring to the appendix where all instances of hedges and boosters are coded in two longer stretches of text.

Hedges and boosters therefore draw attention to the fact that statements don't just communicate ideas, but also the writer's attitude to them and to readers (Halliday, 1978). In the above examples writers are weighting the level of their commitment depending on the epistemic status of propositions as accredited facts or interpretations, and on the anticipated effect this commitment is likely to have on readers' responses. These considerations are an important dimension of academic discourse and a principal way that writers can use language flexibly to adopt positions, express points of view and signal allegiances. They also represent a major contribution to the social negotiation of knowledge and writers' efforts to persuade readers of the correctness of their claims, helping them to gain community acceptance for their work as a contribution to disciplinary scholarship and knowledge.

Research in the social construction of knowledge has clearly demonstrated that knowledge is a cultural product, shaped by the practices of discourse communities and constituted, not just conveyed, by rhetoric (eg. Kuhn, 1970; Rorty, 1979). Academics negotiate the status of their knowledge claims with their peers through the medium of research articles, and success is at least partly dependent on

their use of appropriate rhetorical and interactive elements. Readers accept statements not only because they believe them to represent independent truths or faultless logic, but because they have been persuaded by a writer's systematic appeal to the disciplinary meanings and values that they hold (Bazerman, 1988; Bizzell, 1992; Bruffee, 1986). Writers, then, must socially mediate their arguments, shaping their evidence, observations, data, and flashes of insight into the patterns of inquiry and knowledge valued by their community. In so doing, they are also simultaneously negotiating a harmonious relationship with their readers, framing their arguments in conformity with disciplinary expectations concerning appropriate writer involvement and interpersonal conduct.

Hedges are one of the most studied features of this audience-oriented aspect of claim design. Myers (1989) has suggested that academic writers employ hedges to minimise the potential threat new claims make on other researchers by soliciting acceptance and challenging their own work. Equally however, engagement in disciplinary forums involves norms of interpersonal behaviour underpinned by the sanctions inherent in a system of academic recognition and rewards which hinges on publication (Hyland, 1997). Writers may thus find it easier to satisfy disciplinary gatekeepers when negotiating peer review procedures by observing community expectations concerning collegial deference and limits on self-assurance.

In addition to softening interpersonal imposition, moreover, hedges have also been seen as a way of anticipating the possible negative consequences of overstatement and the eventual overthrow of a claim (Hyland, 1996b & 1998a; Nash, 1990; Salager-Meyer, 1994). By limiting their commitment with hedges, writers offer an assessment of the status of a claim, attesting to the degree of precision or reliability that it carries. Hedges imply then, that a statement is based on plausible reasoning rather than certain knowledge, and allow readers the freedom to dispute it.

At first glance, boosters seem to contradict such conciliatory and defensive tactics. They emphasise the force of propositions and display commitment to statements, thereby asserting the writer's conviction and restricting the negotiating space available to the reader. But while an apparently risky

tactic, boosters nevertheless allow writers to strategically engage with colleagues, effecting interpersonal solidarity and membership of a disciplinary in-group. Although they have received little attention in academic writing, boosters are seen to play an important role in creating conversational solidarity (Holmes, 1984 & 1990) and in constructing an authoritative persona in counselling interviews (He, 1993). In science articles Myers (1989) regards intensifying features as positive-politeness devices, enabling writers to assume shared ground with their readers and stress common group membership. Boosters thus allow writers to negotiate the status of their information, helping to establish its perceived truth by strategically presenting it as consensually given.

In sum, hedges and boosters are a response to the potential negatability of claims and an indication of the writer's acknowledgement of disciplinary norms of appropriate argument. They work to balance objective information, subjective evaluation and interpersonal negotiation, and this can be a powerful persuasive factor in gaining acceptance for claims. This account is complicated, however, by the considerable variation in disciplinary knowledge-making practices. The characteristic ways that writers conceptualise problems, generate inquiry, approach their objects of study, and persuade their colleagues are likely to influence the ways they employ hedges and boosters in their discourse. It is this possibility that I explore in this paper.

Data and methods

The data for this study consists of a corpus of published articles together with a series of interviews with members of the relevant discourse communities. This combination of quantitative and qualitative approaches seeks to provide a thicker account of the discursive activities of these particular groups, allowing the target features to be more clearly interpreted as instances of socially situated practice. It should be noted that informants' accounts were obtained as a means of investigating textual features, seeking to provide further evidence for the social nature of discourse and the relations that underlie the construction and interpretation of texts. I have used the spoken material then not as sociological data, but as a means of situating the texts in academics' understandings of their practices, how they perceive what they do when they read and write in their disciplines. Consequently I have not adopted an

'ironic' stance to these accounts in the manner of Latour and Woolgar (1979) nor sought to deconstruct them with systematic critical analysis. So, while acknowledging that they may well be examined in other ways, in this paper academics' reflections are taken at face value as insiders' perspectives on their everyday social and discoursal activities.

The text corpus consists of 56 research articles, one paper from each of seven leading journals in eight disciplines chosen to represent a broad cross-section of academic activity: mechanical engineering (ME), electrical engineering (EE), marketing (Mk), philosophy (Phil), sociology (Soc), applied linguistics (AL), physics (Phy) and microbiology (Bio). The journals were nominated by expert informants as among the leading publications in their fields (appendix 1), and articles were selected at random from current issues, ensuring that only those based on original data were chosen to facilitate a comparison of linguistic features.

The articles were converted into an electronic corpus of 330, 000 words, and were searched for lexical expressions of hedging and boosting using Wordsmith Tools, a text analysis programme. The search was based on a list of 180 items compiled from grammars, dictionaries and earlier studies, particularly Holmes (1988), Hyland (1996a), Hyland and Milton (1997), and Quirk et al (1972), as well as from the most frequent items in the articles themselves. These items are listed in appendix 2. All cases were then examined by two researchers working independently to ensure that they expressed doubt or certainty, which produced a high inter-rater agreement of 0.87 (Kappa).

The interviews were conducted with experienced researcher/writers from each of the disciplines listed above using a semi-structured format (Cohen & Manion, 1985: 293). This method involves a flexible use of open-ended interview prompts closely related to the research objectives, but allows other topics to be followed-up if raised in the discussion. Part of the interviews required participants to respond to features in selected corpus articles as members of the readership for whom the texts were composed, employing their specialist knowledge as community members to interpret meanings, reconstruct possible writer motivations, and to evaluate rhetorical effectiveness. In the second stage of the

interviews informants discussed their own discoursal practices and examined their own published texts, explaining, as far as they could, their reasons for their own choices in specific contexts. In this way I hoped to obtain the perspectives of insiders acting in their authentic disciplinary roles as both consumers and creators of texts, capturing both sides of the practices of negotiation which create meanings.

All interviews were taped and written up as a summary immediately after the session and subsequently returned to several times, occasionally with the assistance of the subject when I was uncertain about what was meant. While I am aware of the possible meaning-constructing effects of interviewing, it seems to be the most effective way of bringing the insider's perspective to the analysis, taking us nearer to a description of cultural practices in terms of its members' understandings. The unstructured nature of the interviews allowed topics to range widely and speakers to elaborate points in considerable detail, but this also meant that the quotes I have selected for illustration often occurred in much longer stretches of talk. It has therefore been difficult to provide a full context or preceding turns for the spoken extracts without inflicting serious and incoherent digressions on my argument and so I present them here as representative soundbites which best reflect insiders' views.

The remainder of the paper examines the information produced by these methods. I will first present the results of the corpus study to emphasise the significance of boosting and hedging in academic writing and the extent of disciplinary variation. Then I will go on to discuss the pragmatic effects of these features using the interview data as a means of interpreting the patterns found in the texts.

Quantitative Results

The quantitative results demonstrate the importance of hedging and boosting in academic writing, with an average of 120 occurrences per paper, about one every 49 words. Table 1 shows that hedges exceeded boosters by nearly 3 to 1, reflecting the critical importance of both distinguishing fact from opinion in academic discourse and the need for claims to be presented provisionally rather than assertively.

Table 1: Hedges and Boosters in academic articles

Category	Totals	Items per 1000 words	Items per paper
Hedges	4787	14.60	85.5
Boosters	1929	5.88	34.5
Totals	6716	20.48	120

Table 2 shows the most frequently occurring devices used to modify statements, revealing the significance of a relatively small number of modal verbs and epistemic verbs, with *may* and *will* accounting for nearly 17% of all devices in the corpus. The most frequent hedges were *may, would*, and *possible*, and the most frequent boosters were *will, show* and *the fact that.* Epistemic verbs such as *suggest, indicate, assume* and *seem* were also heavily used as hedges.

Table 2: Most frequent hedges and boosters in the corpus

Hedges		Boosters	
Device	Frequency	Device	Frequency
may	646	will	483
would	385	the fact that	123
possible(ly)	306	show (that)	117
could	269	it is clear/clearly	86
might	265	actually	70
suggest	258	indeed	68
indicate	141	always	65
seem	138	obvious (ly)	62
assume	128	of course	56

There were substantial differences in the disciplinary results. Table 3 shows the considerable spread in the frequency of features, with philosophy containing almost 4 times as many devices as physics, and wide distinctions between most disciplines in the use of both hedges and boosters. Mitigation exceeded emphasis in all disciplines, but differences were most marked in the marketing papers and least apparent in mechanical engineering and physics. Overall the results suggest a general division between philosophy, marketing, linguistics and sociology on one hand, and physics and engineering on the other, with biology occupying the middle ground. This distinction is dramatically illustrated by the fact that over 70% of all hedges occurred in the humanities/social science papers and they were over twice as frequent in philosophy, marketing and linguistics, as in physics and engineering.

Table 3: Disciplinary differences

	per paper		per 1,000 words		number	
	Hedges	Boosters	Hedges	Boosters	Hedges	Boosters
Philosophy	137.3	72.3	18.5	9.7	961	506
Marketing	136.3	48.3	20.0	7.1	954	338
Applied Linguistics	114.0	39.1	18.0	6.2	800	274
Sociology	96.3	33.4	13.3	4.6	674	234
Biology	78.9	22.6	13.6	3.9	552	158
Electrical Engineering	45.6	17.7	8.2	3.2	319	124
Mechanical Engineering	39.3	20.4	9.6	5.0	275	143
Physics	36.0	21.7	9.6	6.0	252	152

The science and engineering papers were again heavily underrepresented in the number of boosters, although the overall discrepancies were generally not so large. There was more of a disciplinary imbalance here though, with philosophy and marketing papers evidencing considerable use of boosters. Over a quarter of the total boosters occurred in philosophy and less than 7% in electrical engineering.

To summarise, the results show major disciplinary differences in the rhetorical preferences of academic writers. This reveals itself as a clear distinction between the sciences and humanities/ social sciences. Papers in philosophy, sociology, marketing, and linguistics contained about 2 ½ times as many devices overall, with hedges particularly strongly represented.

Disciplinary conventions and discourse choices

Hedges and boosters are interpersonal aspects of language use, complex textual signals by which writers personally intervene into their discourse to evaluate material and engage with readers. Their presence or absence in a text might therefore be seen as the discoursal choices of individual researchers deciding to represent themselves more or less explicitly in their writing, either adopting a clear authorial presence or linguistically suppressing this identity. Clearly these choices are to some extent influenced by individual personality factors, such as self-confidence and experience, and we often regard them as largely unreflective and automatic aspects of writing. However, all acts of communication occur in social contexts and carry the imprint of those contexts. In academic writing,

the choices individuals make are socially shaped and constrained by the possibilities made available to them by the discourse conventions of their disciplines.

I note earlier that one way of approaching these interpersonal conventions has been to regard hedges and boosters as politeness devices employed to smooth the threats said to be inherent in the routine interactions of academic knowledge-making. Building on Brown and Levinson's (1987) model, Myers (1989) suggests that presenting, denying and evaluating claims in academic writing constitute "Face Threatening Acts", or impositions on the self-image of readers, which have to be mitigated by various rational strategies. Because these acts solicit acceptance and supersede the claims of others, they often contradict the literature or challenge the assumptions underlying others' research. Hedges repair such potential threats by emphasising the provisionality of statements, while boosters stress shared wants and concerns. While this is an insightful and exciting theory, it nevertheless underplays the importance of disciplinary norms in disciplinary practice, and the critical interdependence of the system of academic communication and material rewards. Essentially the theory neglects the exercise of power and conformity in the discourse culture, for while writers weigh up their rhetorical choices and the potential effects of their statements, they do so with the awareness that publication, reputation, and career may ride on the outcome.

Most centrally, these discourse conventions embody the particular sets of values, practices and beliefs which are held by, and help define, academic disciplines. An example is the obvious emphasis that academe places on the need for care and prudence when weighing evidence and drawing conclusions from data. In all disciplines, making an appropriate level of claim for one's findings is a critical aspect of research, and writers are expected to evaluate their propositions as accurately and objectively as possible. This model of scholarly objectivity and exactitude is often seen as a particular virtue of the physical sciences, a view reflected in these comments by my respondents:

> *In science it's a bit harder than the arts I suspect. You are always cautious. You can never guarantee certainty.* (EE interview)

In my field rhetoric is important, unlike the sciences. But I like to think that the way I do research is scientific and my attitude is a scientific one. To be more cautious than assertive. (Mk interview)

With biological systems you are always not quite sure. Most of the time you could be right, but there is always a chance it might be something different. (Bio interview)

There are always doubts even though findings might point a certain way. As a journal editor I've often had to insist on authors reducing their claims if the evidence doesn't support it. (Phy interview)

Circumspection and discretion are clearly important constraints in formulating knowledge claims in the sciences, and this may help account for their low use of boosters. However, the fact that these devices occurred in all the papers in my corpus, and that hedges were far more frequent in the social sciences/humanities, testifies to a more complex rhetorical picture. As I have noted, no absolute distinction can be drawn between 'fact' and 'evaluation' because claims are interpreted through a prism of disciplinary assumptions. Readers bring certain expectations of exactitude to a text and writers attempt to meet these. So, while my results perhaps coincide with the popular intuitions that the sciences tend to produce more impersonal texts, or that they have more rigorous standards of precision, there is clearly more to it. Essentially the findings reflect the fact that research articles are manifestations of the different epistemological and social assumptions of disciplinary communities. Writers present their work in different ways partly because they have different sorts of work to present (Nash, 1990), but also because they are presenting it to people with different ways of seeing and describing the world.

In other words, because academic writing is a form of knowledge-making, differences in the types of problems studied and ways of addressing them should help account for disciplinary variations. I want to suggest, then, that the regularities identified here offer insights into the knowledge constructing procedures of disciplinary communities. In particular, I will try and show that the ways writers modify the strength of their statements broadly reflect the types of intellectual inquiry and knowledge structures peculiar to the hard and soft disciplines.

Knowledge contexts and knowledge claims

The concept of hard and soft domains of knowledge is obviously not without problems, partly because it is an everyday term which carries connotations of clear-cut antithetical divisions. As a result, using it to characterise academic disciplines by types of knowledge forms clearly runs the risk of reductionism, or even reification, by packing a multitude of complex abstractions into a few simple opposites. Moreover, for some the terms may seem ideologically loaded, privileging a particular mode of knowing based on the structural perspectives, symbolic representations and model building methods of the natural sciences. However, the hard-soft scheme is more directly related to established disciplinary groupings than some more abstract categorisations (Becher, 1989). Moreover, evidence from a questionnaire survey of academics (Biglan, 1973) and from psychometric tests of students' learning strategy preferences (Kolb, 1981) suggest that it may actually represent actors' own perceptions of the areas in which they are engaged. If the hard-soft distinction is conceived as a continuum, rather than as unidimensional scales, I believe it offers a convenient way of examining general similarities and differences between fields without positing rigidly demarcated categories.

One distinction between hard and soft domains is that scientists generally see their knowledge as coming from relatively steady cumulative growth. Problems emerge from prior problems and there are fairly clear-cut criteria of what constitutes a new contribution and how it builds on what has come before (Becher, 1989; Hyland, 1998b). When interview discussions turned to epistemological issues, my informants needed no encouragement in asserting their endorsement of this position:

> *My personal view of science is that of a huge volcano and lava is flowing down and I'm at the end of one stream of lava.* (Phy interview)

> *There are many groups making infinitely small steps forward on a particular problem, eventually someone may make a bigger step and get a Nobel Prize, but if not, the groups will get there anyway.* (EE interview)

This representation may, of course, merely reflect the epistemological ideology of practising scientists, but it nevertheless has practical and rhetorical effects. One consequence is that many scientists and

engineers typically work in what Becher calls "urban research communities", where large numbers of researchers work on relatively few problems. This means issues are clearly defined and schools of researchers have similar understandings to draw on, writers can therefore presuppose a certain amount of background knowledge, procedural expertise, and technical lexis.

The soft-knowledge areas however are typically more interpretative and less abstract. Research is often influenced by contextual vagaries, there is less control of variables and more diversity of research outcomes. Writers frequently draw on out-of-discipline research and there are fewer unequivocal bases for accepting claims. With the exception of philosophy, readers in the soft disciplines are themselves often more heterogeneous, with different academic or professional backgrounds and more varied purposes in reading. Again, my informants stated that they considered this in their writing:

> *I like to think that both professionals and academics read my work, I want to make an impact in the workplace so I write with this in mind. Its not just for academics.*
>
> (Mk interview)
>
> *I'm bringing in stuff from composition theory, cognitive psychology, and sociology that may not be familiar to linguists.* (AL interview)

Together this means that writers in the soft fields can generally take less for granted and, while a paper must carry conviction, it must also appeal more to the reader's willingness to follow the writer's reasoning. Research cannot be reported with the same confidence of shared assumptions and so has to be expressed more cautiously, using more hedges. Writers must rely far more on focusing readers on the claim-making negotiations of the discourse community, the arguments themselves, rather than relatively unmediated real-world phenomena (cf. MacDonald, 1994). These typical examples from my linguistics corpus give some flavour of this:

> (5) If it can be demonstrated that the rejection of the Subjacency violation may possibly be the result of factors other than UG, then certainly such a conclusion is unwarranted.
>
> However, it seems likely that the context in which these students study is important in understanding the results.

We tentatively suggest that The Sun's minimalist style creates an impression of working-class language, or restricted code, while the very wordy Times themes, especially their long qualifiers and apposition elements, remind one of academic, formal discourse

This might be because, in China, English language learning in middle schools and in university is very different in terms of goals, methods, and learning conditions.

Differences between the hard and soft domains were not only apparent in the large disparity of hedges, but also in the type of hedges used. In particular, writers in the hard-knowledge fields made over twice as much use of attribute hedges (Hyland, 1996a, 1998a). These are devices like *about, approximately, partially, generally, quite,* and so on which differ from other hedges in that they refer to the relationship between propositional elements rather than the relationship between a proposition and a writer. Thus such devices limit the scope of the accompanying statement, rather than cast doubt on its certainty:

(6) Heat dissipation can be largely attributed to line sources and has a time constant rTH, typically in the region of a microsecond. (EE)

Because of transportation costs, domestic consumption for metallurgical coal is usually met by imports from the United States, while virtually all domestic metallurgical coal production is sold on the export market. (ME)

Although our data generally support these former results, factors other than T-DNA copy number are clearly involved (Bio)

Precursor CWNMRON provides independent information on 4 the Larmor inhomogeneous broadening, although in practice the pulsed NMRON linewidths were somewhat broader than the low power CWNMRON values. (Phy)

In these examples writers are not using hedges to dilute their certainty or withhold commitment to their propositions. Instead they are seeking to present a situation in terms of how far it varies from the ways the discourse community conventionally sees the world, either restricting the temporal or qualitative range of the claim or its generalisability. Attribute hedges therefore indicate the extent to which results fit a standard disciplinary schema of what the world is thought to be like, signalling a departure from commonly assumed prototypicality.

Attribute hedges are an important component of knowledge contexts and the kinds of claims these allow because they directly signal an appeal to such contexts by invoking the cultural understandings of participants. Attribute hedges do not only suggest a certain amount of community agreement on

what might reasonably be expected, they also draw on shared standards of permissible imprecision (Channell, 1990; Dubois, 1987). Grice's maxim of quantity states that writers will present data accurately enough for the purposes they serve, suggesting that choices are likely to be contextually variable. In academic writing, and particularly when used with numerical data, attribute hedges allow writers to draw on unspoken conventions of imprecision.

Clearly writers will always seek to present data in ways most favourable to their argument, and by employing these devices they are able to rhetorically appeal to their readers' specialised shared knowledge regarding what can be safely taken for granted as background. This is because attribute hedges rely on the reader's recognition that the information presented at this point is non-significant, these shared understandings providing the writer with a warrant to rhetorically subordinate the data in order to highlight more central, and more precise, quantities elsewhere. These examples are typical:

(7) With winter acclimation, the LT50 (temperature for 50% leaf death) appears to be about -5 °C at our warm site and about -6 °C at our cold site. (Bio)

Membrane-based circuits are typically large (or long) because the effective dielectric constant is close to 1.0. (EE)

A simple calculation, assuming no positive interaction between the polymer molecule and the interface [29], shows that under these conditions approximately 0.5% of the molecular segments are within a distance that can conceivably interact with the substrate. (Phy)

The budget of commercial lighting programs often makes up 25 percent or more of the total DSM expenditures of the utilities. (ME)

The greater use of attribute hedges found in the hard science papers therefore suggest that writers make more use of shared understandings to either indicate variability or conventions of imprecision. This rhetorical feature therefore points to a more cohesive body of consensual knowledge than typically appealed to in the soft fields.

Authorial involvement in knowledge construction

Another possible explanation for the wide differences in the use of hedges and boosters between different disciplines is that the distribution of these forms embody very different assumptions about the role of human actors in the construction of knowledge. In this section I will briefly outline what I

see as the main issues that the use of hedges and boosters raise in this regard: (i) writer presence, (ii) subjectivity, (iii) interpersonal engagement, and (iv) writer commitment.

1. Minimising writer presence

An important aspect of the positivist-empirical epistemology typically found in the hard sciences is that the authority of the individual is subordinate to the authority of the text. Writers generally seek to disguise both their interpretative responsibilities and rhetorical identities behind a screen of linguistic objectivity. A prudent scientist avoids using features which reveals either a personal involvement in the rendition of findings or a commitment to that reading. Lab experiments are believed to produce accurate depictions of the real world, and their textual representation are best designed to be faceless and agentless, claiming an appearance of objectivity and neutrality (Hyland, forthcoming).

Impersonalisation strategies such as the use of passives, nominalisation, and objective theme selections have been well-documented in the literature and appear to represent the rhetorical face of science (eg. Gosden, 1993; Halliday, 1988; Swales, 1990). Together these features help reinforce the predominant view of science as an impersonal, inductive enterprise. In other words, they contribute to the ideological representation that scientists discover truth rather than construct it, minimising the role of socially contingent factors in scientific research practices (cf. Gilbert & Mulkay, 1984; Latour and Woolgar, 1979).

Clearly the less frequent use of hedges and boosters is one more way of minimising the researcher's role in interpreting data, evaluating claims, and appealing to readers. By emphasising writer invisibility, they appear to allow the facts to more transparently speak for themselves. This effect is often reinforced by the use of an embedded clause with an anticipatory or dummy *it* replacing a human agent as subject:

> (8) <u>It is evident that</u> the enzyme and its rnRNA are present in high concentrations throughout the degradation of the old clove and the formation of the new bulbs. (Bio)
>
> <u>It is possible that</u> an increase in ethylene production in these fruits, a vital physiological event during ripening, is mediated by CABA. (Bio)

Although the error increases when `1/1 is less than 0.01 or larger than 1.0, it seems that the ratio d/1 in the range from 0.01 to 1.0 gives accurate, stable results. (ME)

It was suggested that rays be returned from the eye's lens, at angles close to the Brewster angle, which would in fact constitute a retrodirector (as defined below). (Phy)

It was shown that the algorithm can function well during actual welding when various noises are present. (EE)

In addition, the science/engineering papers contained a higher proportion of discourse-oriented verbs like *indicate, suggest* and *show* among their hedges and boosters. These devices carry less subjective connotations than cognition verbs such as *think, believe* and *suspect*, and are also more easily combined with inanimate subjects. This allows writers to more easily attribute agency to "abstract rhetors", a common practice in my science and engineering data:

(9) X-ray diffraction analysis shows that the composite consists of xSiNa and siN from the matrix and SiC from the Hi-Nicalon fibres. (Phy)

The results presented here clearly indicate that alliinase and the leaf and bulb lectins are differently regulated during the life cycle of the garlic plant. (Bio)

Figure 7 demonstrates the degree to which heat transfer varies during combustor warm-up. (ME)

This figure indicates clearly that the energy absorption levels for all three planes for a concave joint are higher than those for a convex joint. (ME)

The data suggest that the saturation point of our core material limits the maximum magnetic flux density. (EE)

These findings show that choosing Kanban parameters solely based on the mean can result in a poor design. (EE)

So here the decision to foreground the "X-ray diffraction analysis", for example, presents a view where a research entity takes responsibility for the asserted proposition and minimises the interpretative role of the researcher.

This practice of giving prominence to procedures or data, rather than themselves, when drawing inferences was recognised by my informants in their own work:

Its conventional to use these formulas to keep yourself out of the picture. They are just conventional ways of expressing inference. (ME interview)

We rely very much on statistical appraisal of results to be able to say something is happening or not, but the big difficulty is making a causative link. Generally I think we'd prefer to say the relationship lies in the data than our heads. (Bio interview)

Of course, I make decisions about the findings I have, but it is more convincing to tie them closely to the results. (Phy interview)

You have to relate what you say to your colleagues and we don't encourage people to go out and nail their colours to the mast as maybe they don't get it published.

(Bio interview)

Clearly there are different reasons why writers may seek to distance themselves from their interpretations of data (Hyland, 1998a), but the net effect is the suppression of the author's voice and the creation of a discourse where the research appears to speak for itself.

2. Subjectivity and soft knowledge

The strategic management of evidentiality and affect appears to work quite differently in the soft disciplines. Not only did the papers in philosophy, marketing, linguistics and sociology contain almost 1½ times the number of hedges and 3 times the boosters, but writers in the soft disciplines were also more likely to stress subjectivity when modifying statements. This is partly conveyed by a more frequent use of cognition verbs, which carry a greater sense of personal conjecture to the modified statement:

(10) This stems, I believe, from the fact that lack of ignorance is a root requirement for responsibility. … (Phil)

Although further research is needed, we suspect that the type of new product used in this study (ie., one designed to enable people to take medicine more easily) may have contributed to this result. (Mk)

This limited perspective is troubling to us in that we feel it could lead to inadequate theories of composition and consequently to instructional practices that are ineffective or even counter productive. (AL)

As far as I know, this account has gone unchallenged. (Phil)

We hypothesize that desires, like expectations, will affect disconfirmation negatively. (Mk)

In addition, as suggested by the above examples, these writers were also more likely to accept personal responsibility and more strongly convey their involvement by the use of personal attribution. Both

hedges and boosters co-occurred far more with first person subjects in the soft fields as writers sought to position themselves in relation to their work:

> (11) We demonstrate that a customer orientation is particularly useful to marketing all innovation in environments in which demand is uncertain. (MK)
>
> Those principles, so I claim, apply to the choices that people make within the legally coercive structures to which, so everyone would agree, principles of justice (also) apply. (Phil)
>
> The explanation we consider most probable is that the matriculation examination, designed for a population far more diverse in English proficiency than GEM4, would not have discriminated as effectively as GEM4 amongst this relatively homogeneous, high-ability group, thus reducing its predictive effect. (AL)
>
> I show why the experiences of very cognitively disabled people should be a central concern of critical sociologists, social historians and social theorists. (Soc)

My sociology informant observed this was a conscious choice for him, related to his perception of self as a writer and his relationship to his discipline:

> *I'm very much aware that I'm building a façade of authority when I write, I really like to get behind my work and get it out there. Strong. Committed. That's the voice I'm trying to promote, even when I'm uncertain I want to be behind what I say.* (Soc interview)

3. Participant relationships and interpersonal engagement

While the extracts presented above show writers explicitly indicating a personal attitude to their findings or positions, hedges and boosters in the soft disciplines also often carried a strong interpersonal element. Hedges can function as a resource for structuring a relationship between participants and accomplishing a more receptive reader attitude to claims. By employing markers of evidentiality with inclusive pronouns, for instance, writers are able to construct a shared context with their readers and draw on assumed beliefs specific to their particular social group. In this way, the writer can signal the status of the information as possible, given what interlocutors might appropriately assume as rational colleagues. This encourages the reader to participate as an intelligent equal in the reasoning process:

> (12) Given that 'style' points to a given mode of consumption, one may suggest that discussing the moral and social merits of a given sexual life-style is in a sense more evidently subjective and open-ended than arguing that the criteria for 'appropriate' sexuality is its (re)productivity. (Soc)

One could conceivably conclude from this type of result that the subjects both 1) have different knowledge representations concerning the L2 and 2) this difference is manifested in online processing tasks. (AL)

We seem to have here a kind of de dicto-de re ambiguity in the verbal form of (3) and without benefit of any intensional operator. (Phil)

One can, of course, speculate about what might likely be causing the difficulty which in each case generates the repair initiation. (AL)

Consequently, one may speculate that, given the relative power advantage of suppliers over ED resellers, one-way flow of information from suppliers to ED resellers would dominate the communication activities in ED channels of distribution. (Mk)

Similarly, the writer may explicitly solicit support for particular assumptions or conclusions using boosters. Boosters can assist the writer's persuasive intentions in a variety of ways, but interpersonally they function to mark, or rhetorically manipulate, consensual understandings based on shared community membership. Devices such as *of course* and *obviously* are often used to signal the writer's assumption that a proposition carries the status of mutual pre-existing knowledge, and the effect of this is to strategically align one's claim with the knowledge of the field:

These are finesse words. I suppose I'm gently leading the reader to my view. (AL interview)

I don't like blunt assertions, but I'll use words like this if I think an idea is sound and that there are good reasons people should go along with me. (Mk interview)

By including readers in this way, the writer credits them with possessing both in-group understandings and the intelligence to make the same reasonable inferences. The argument is thereby strengthened by claiming solidarity with the community and the mutual experiences needed to draw the same conclusions as the writer:

(13) Assuming, of course, that a GJ task actually reveals something about a subject's underlying linguistic competence. (AL)

This conclusion would obviously suggest a massive impairment to the language learning capacity. (AL)

One obvious implication of the increased environmental awareness is that manufacturers will have to take more care to use environmentally friendly packaging and propellant formats. (Mk)

Foucault's well-known thesis is that sexuality became an issue of crucial importance from the eighteenth century onward because it helped combine the control of two important entities: an individual body and the population. … (Soc)

Reader relationships are particularly evident in philosophy where effective argument relies heavily on point scoring, humour and positive face engagement (Bloor, 1996). Here the 'result' of an argument is not physical data or human behaviour, but the argument itself. Writers typically seek to manoeuvre both supporters and opponents into agreement with their position by using strategies which employ a degree of conventional intimacy unknown in other disciplines. One way of creating this sense of solidarity is by using boosters to appeal to the reader as an intelligent co-player in a close-knit group. These examples are taken from my philosophy corpus:

> (14) By now the moral should be obvious. It is clearly futile to search for something for which Perry might be directly culpable but concerning whose wrongness he was ignorant.
>
> However, it is clearly defective in some way.
>
> So it looks as if indicative conditionals must be material implications.
>
> Furthermore, the argument for A-entailment is not just that it is obvious, as I show below. There is a theoretical ground for it.
>
> Pomerance is just wrong, however.

Thus, sharing in-group knowledge and flattering readers' intelligence is commonplace in philosophy and this high degree of personal involvement helps account for the fact there were almost twice as many boosters in philosophy as any other discipline except marketing.

4. Commitment and propositional force

Boosters were also relatively common in the physics, marketing and applied linguistics papers, but here their primary role was not overtly interpersonal. Instead, they appeared to support the writer's argument in one of two ways. First, and mainly in the soft disciplines, they served to emphasise the strength of the writer's commitment to a proposition, and thereby sought to convince the reader by their belief in the logical force of the argument. Two comments from informants exemplify this view:

> *You have to be seen to believe what you say. That they are your arguments. Its what gives you credibility. It's the whole point.* (Phil interview)

> *I like tough minded verbs like 'think'. It's important to show where you stand. The people who are best known have staked out the extreme positions. The people who sit in the middle and use words like 'suggest', no one knows their work.* (Soc interview)

This kind of commitment is conveyed in the extracts in (15), where a variety of boosters are used to carry the writers' authority:

(15) But it is also evident, and this is the central point here, that in initiating repair on Margy's response, Emma is treating that response as not properly fitted - as an inappropriate response - to the action which she intended her turn to be heard as doing. (AL)

Further research should definitely explore this point. (AL)

It is certainly meaningful to refer to street homeless people as 'going to work', in the sense that they engage in regular activities which 'earn' them sufficient to meet their daily subsistence needs. (Soc)

It is indeed the case that the same more inclusive standards of political citizenship have been extended not only to women but also to children, and arguably, to fetuses. (Soc)

This particular result is undoubtedly attributable to the impending incorporation of Hong Kong into the People's Republic of China. (Mk)

As we have seen, this, quite simply, is not so. (Mk)

The second way that writers employed boosters was to comment impersonally on the validity of their propositions. Boosters were either used to stress the strength of warrants, suggesting the efficacy of the relationship between data and claims with verbs such as *establish* and *show*, or express the certainty of expected outcomes, often with *predict* and *will*:

(16) Consequently it is established that such a configuration constitutes a retrodirector, rather than a simple reflector. (Phy)

Numeric model results demonstrate that the best pressure gain performance is indeed achieved when these frequencies match. (ME)

The results offer clear support for the proposed model. (Mk)

Listening will continue to play a large part in pronunciation training, with perhaps more authentic listening tasks with a variety of accents. (AL)

On the basis of our discussion, we predict that CA associations influence the perception of important product attributes (i.e., product sophistication in our empirical test). (Mk)

However, once the hydrostatic stress becomes tensile, voids that have nucleated will grow and eventually coalesce to form an extrusion defect. (ME)

The main disciplinary distinctions however, once again involved a preference for impersonal strategies in the hard sciences. In the science and engineering papers a higher proportion of hedges and boosters were modal verbs, which are less specific in attributing a source to a viewpoint. That is, modals tend to downplay the person making the evaluation:

(17) The slice at w = w, will give the useful information. (Phy)

From the AHk values measured here, it may be concluded that Gd" ions prove to be weak relaxing ions of the rare-earth series. (Phy)

The theory given above simply provided some insight into the various mechanisms and configurations that might or might not yield a polarimetric effect. (Phy)

There was a good correlation between the four values. For V. trifidum, ANOVA (Fig showed a significant increase from L to L' and FI, which could be interpreted as reflecting the dynamics of fungal colonization. (Bio)

A counter-clockwise hysteresis was observed in the C-V curve of sample A (Fig. 2). It is well known that the direction of the hysteresis loop from the MIS capacitor due to surface-state trapping should be clockwise[9]. The motion of charges in the insulator should thus play a major role in this hysteresis effect. (EE)

For reasons that will become apparent in this paper, it is expected that external hardware could have a significant effect on the optimal performance of a pressure gain combustor. (ME)

These distributions can be seen in the longer extracts reproduced in the appendix. These samples are fairly typical of the discussion sections of biology and marketing papers in the corpus as a whole, although the biology excerpt was selected for illustrative purposes and contains rather more hedges than is usual. The marketing extract contains nine boosters and 25 hedges, and biology six boosters and 36 hedges. In both extracts boosters (boldface) are heavily outnumbered by hedges (underlined) and demonstrate far less varied lexical realisations, largely restricted to *show, demonstrate* and *find*. One difference is that boosters are often employed in the marketing paper to highlight the significance of the research, but only by the biologists to accent firm conclusions. In both extracts the more assertive claims are largely restricted to specific experimental results, and these are themselves often subsequently hedged.

Hedges are employed when the writers move away from what can be safely assumed or experimentally demonstrated, but there are important differences in how these are used in the two samples. In the biology example almost all devices (95%) are expressed impersonally, with abstract subjects, deleted agent passives, and a heavy use of dummy ‘it’ subjects, such as ‘it seems unlikely’, ‘it has been shown’, and ‘it is generally believed’. While similar forms are also to be found in the marketing paper, 28% of epistemic devices were framed using personal subjects, particularly ‘we

found' and 'our results suggest'. Modals occur almost equally in the two samples, with *could* predominating in biology and *may* in marketing, although attribute hedges are far more frequent in biology, with forms such as *usually, virtually, about,* and *as much as* acting to limit the efficacy of a possible explanation, or functioning to fit the vagaries of experimental results into a disciplinary schema. So, while marketing and biology are perhaps not at polar ends of the hard-soft continuum, it is nevertheless possible to identify some characteristic patterns of these larger dimensions of disciplinary practice.

To summarise the overall discussion of my results, writers in the soft fields relied more on a personal projection, while scientists and engineers tried to portray their evaluations impersonally, constructing a context in which claims appeared to arise from the research itself. Once again, however, there are good reasons for understanding these disciplinary preferences for mitigation and assertion not merely as obedience to arbitrary conventions, but as rational attempts to make the best use of linguistic resources to effectively interact with colleagues and secure agreement for one's arguments.

Conclusion

In this paper I have explored some of the contextual factors which shape the ways we say what we know, and want others to accept. I have tried to show that the expression of doubt and certainty is central to the negotiation of claims, and that what counts as effective persuasion is influenced by different epistemological assumptions and permissible criteria of justification. My own use of hedges and boosters in this paper, for example, have been shaped by an awareness of the need to temper personal conviction with community practice. Any success I may have had in persuading you of my claims then is, at least in part, a consequence of you granting that my commitment to them has been appropriately managed.

This study is a contribution to the growing literature which contends that the features of academic texts can only be fully explained when considered as the actions of socially situated writers. While individual factors doubtless contribute to the choices made by particular writers in the moment-by-moment creation of a research paper, meanings are ultimately produced in the interaction between

writers and readers in specific social circumstances. Writing as a physicist, linguist or an engineer means being able to talk to your peers in ways they are likely to find convincing. In other words, what we see on the page as an author's decision to intervene to urge acceptance, solicit solidarity or counsel caution, or as an option to step back and encourage cold judgement, reflects clear interactional and institutional understandings.

Although I have only sought to offer a broad characterisation of hedges and boosters in academic articles, it is apparent from the analysis that these are complex devices with sophisticated functions. The distributional patterns however do not only emphasise the importance of these features and their wide disciplinary variation. They also underline the fact that future research in this area must consistently take account of both contextual and quantitative factors if the results are to be useful. It is clear that the use of hedges and boosters in academic discourse is regulated both by general rules of communication and the norms and practices of particular disciplines. But further research is needed to extend this study into other disciplines and genres, and into the use of other discoursal features. In this way we might hope to gain both a greater understanding of the resources academic writers employ for constructing a sympathetic reader environment for their claims, and of the ways that academic cultures can be regarded as distinctive.

Appendix 1: Journal Corpus

Applied Linguistics

Applied Linguistics
TESOL Quarterly
Second Language research
System
English for Specific Purposes
World Englishes
Journal of Second Language Writing

Marketing

Journal of Marketing Management
International Journal of Research in Marketing
Journal of Marketing Research
Journal of Marketing
Journal of the Academy of Marketing Science
Journal of Marketing Communication
Marketing science

Philosophy

Mind
The Journal of Philosophy
Analysis
The Philosophical Quarterly
Philosophy
Erkenntnis
Inquiry

Sociology

American Journal of Sociology
The Sociological Review
Current Sociology
International Journal of Comparative Sociology
Sociology
International Sociology
British Journal of Sociology

Physics (Magnetics)

Journal of Magnetism and magnetic materials
Bulletin of Magnetic Resonance
Applied Magnetic Resonance
Electromagnetics
Journal of Magnetic Resonance
J of Electromagnetic Waves and Applications
Journal of Magnetic Resonance

Electrical Engineering

Microelectronics Journal;
Microsystem Technologies
Microwave Theory and Techniques
Solid State Electronics
Analog Integrated Circuits & Signal Processing
International Journal of Microwave and Millimeter-wave Computer-aided Engineering
Journal of Micro-electromechanical Systems

Mechanical Engineering

Mechanism and Machine Theory
Mechanics and Material Engineering
Int Journal of Mechanical Sciences
Journal of Process Mechanical Engineering
Journal of Mechanical Engineering Science
Journal of Engineering Manufacture
Energy Sources

Microbiology (cell biology)

Journal of Cell Biology
Mycological Research
The Plant Cell
Plant Molecular Biology
Plant, Cell and Environment
Molecular and Cellular Biology
Mycologia

Appendix 2: List of Hedges and Boosters examined in the study

about	(of) course	indeed	possible	superficially
actually	decidedly	indicate	possibly	suppose
admittedly	deduce	inevitable	postulate	sure
almost	definitely	infer	precisely	surely
always	demonstrate	interpret	predict	surmise
always	determine	(we) know	prediction	suspect
apparent	discern	(it is) known	predominantly	technically
apparently	doubt	largely	presumably	tend
appear	doubtless	least	presume	tendency
approximately	(is) essential	likely	probability	theoretically
argue	essentially	mainly	probable	(we) think
around	establish	manifest (ly)	probably	true
assume	estimate	may	propose	typically
assumption	evidence	maybe	prove	unambiguously
assuredly	evident	might	provided that	unarguably
basically	evidently	more or less	(open to) question	uncertain
my/our belief	expect	most	questionable	unclear
I believe	the fact (that)	must	quite	undeniably
certain extent	we find	necessarily	rare	undoubtedly
certain that	formally	(not) necessarily	rarely	unequivocal
certainly	frequently	never	rather	unlikely
certainty	general	no/beyond doubt	relatively	unmistakably
claim	generally	normally	reportedly	unquestionably
(to be) clear	given that	obvious	reputedly	unsure
clearly	guess	obviously	seems	usually
conceivably	however	occasionally	seemingly	virtually
conclude	hypothesize	often	(can be) seen	well-known
conclusive	hypothetically	ostensibly	seldom	will
confirm	ideally	partially	(general) sense	won't
conjecture	implication	partly	should	would
consistent with	imply	patently	show	wouldn’t
contention	impossible	perceive	sometimes	wrong (ly)
convincing (ly)	improbable	perhaps	somewhat	
could	inconceivable	plausible	speculate	
couldn't	incontrovertible	possibility	suggest	

Appendix 3: Coded features from Discussion section of a Biology paper (930 words)

Note: **Bold** = Booster; Underline = Hedge

DISCUSSION

Despite the widespread occurrence of CABA in higher plants, its biological role(s) is still not clearly elucidated. Several possible roles of CABA have been suggested. GABA may be involved in cytoplasmic pH regulation (refs), temporary nitrogen storage, defense mechanisms against pests and disease (refs), and nitrogen transport (ref). **The work presented in this paper demonstrates** that CABA can promote synthesis of the 185 rRNA in C is about 1.8 kb. when CABA was added to ACC oxidase extracts, there was no detectable production of ethylene. Therefore, CABA is probably not acting as a direct ethylene biosynthetic precursor.

ACC synthase is often considered as the rate-limiting step in ethylene biosynthesis (ref). When ACC synthase was inhibited by AVG, CABA failed to promote ethylene synthesis, indicating that CABA stimulates ethylene by acting on ACC synthase. Consistent with this idea, endogenous ACC levels were severalfold higher in CABA treated tissues. This result also suggests that there is a higher ACC synthase activity in vivo in GABA-treated tissues. Such an increase in ACC synthase activity could be due to de novo protein synthesis, as judged by the abundance of ACC synthase transcripts in CABA-treated tissues. When the tissues were supplied with both AIBA and GABA, the effect of GABA on ethylene was virtually eliminated, suggesting that CABA could have enhanced ethylene biosynthesis only via ACC.

Even though ACC oxidase is usually considered as a constitutive enzyme (ref), a growing number of recent studies indicate that ACC oxidase can also be regulated (refs) and thus be involved in regulation of ethylene biosynthesis. In vitro ACC oxidase assays done in our study **show** an increase in ACC oxidase activity in GABA treated tissues. A parallel increase in accumulation of ACC oxidase transcripts in GABA-treated tissues was also observed. The increase in ACC oxidase MRNA abundance and enzyme activity could be due to GABA itself and/ or a possible positive feedback by the ethylene induced by GABA.

It seems unlikely that GABA imposes a chemical stress, since the endogenous GABA levels in GABA-treated tissues show an increase of only 7-fold after 12 h, indicating that either less GABA is taken up than might be expected or that GABA is taken up and is quickly utilized by tissues. Similar to our finding, Aurisano et al. (1995) reported that rice seedlings accumulate as much as 8 limol GABA/g fresh weight under anoxic conditions. These authors, however, observed only about 1 btmol GABA/g fresh weight after soaking the tissues in 1 mm GABA. This may also partly explain why large GABA doses were required to increase ethylene production in our study. Yet one cannot completely rule out the possibility that GABA induced the ethylene production by exerting a chemical stress. For instance, CABA could have indirectly altered the cytoplasmic pH status (ref) or incorporated itself into proteins by mimicking protein amino acids, which could have been perceived by plant cells as stress. Stimulation of ethylene production by D-iSomers of several amino acids has been reported by Satoh and Esashi (1980b) in various cocklebur seed tissues. Similarly, Cohen et al. (1994) reported that a- and P-aminobutyric acid, but not CABA, can stimulate ethylene production in tomato plants. Thus, our results suggest that the ability of CABA to stimulate ethylene production might differ in various plant species.

Recently, **it has been shown that** over-expression of active GAD causes severe abnormalities in growth and development of tobacco plants (ref). In the present study the observation that ethylene production is stimulated only by GABA but not by glutamate, the precursor of CABA, indicates the importance of regulation of GAD activity in plants. GABA is usually shunted back to the Krebs cycle in the form of succinate. Since succinate could not increase ethylene production, it is unlikely that CABA enhanced ethylene synthesis by increasing the flux through the Krebs cycle. Rather, CABA may act as a signaling regulatory molecule and cause ACC synthase transcript accumulation at transcriptional and/ or post-transcriptional levels. For instance, the rapid 10- to 25-fold increase in endogenous GABA levels following mechanical damage (ref) can stimulate ethylene production in a manner **shown** in the present study.

Hypoxic conditions cause a sequential increase in CABA (ref) and ethylene levels (ref). **It is generally believed** that the decline in cytosolic pH during hypoxia might trigger the GAD

activity that leads to GABA accumulation (ref). The increase in ethylene production that is seen in certain tissues when roots are under hypoxic conditions is caused by enhanced ACC synthase activity (ref). Low 0, concentration is considered as a signal for inducing ACC synthase (ref). Perhaps this signaling is in part aided by the increased GABA levels.

The other instance in which GABA might mediate ethylene biosynthesis is during ripening of climacteric fruits. Inaba et al. (1980) reported that CABA and Clu are the most abundant amino acids in tomato fruits. T e GABA levels were higher until the late breaker stage (pink) of ripening tomato fruits. Gallego et al. (1995) recently reported a corresponding rise in the accumulation of a calmodulin-binding GAD transcript during the onset of ripening in tomato fruits. It is possible that an increase in ethylene production in these fruits, a vital physiological event during ripening, is mediated by CABA.

In an effort to broaden our understanding of the roles of CABA in plants, we investigated the possibility that there might be a role for GABA in the ethylene biosynthetic pathway. **We have shown** that GABA does not appear to be an alternative biosynthetic precursor for ethylene. CABA, however, enhances ethylene synthesis by promoting the accumulation of ACC synthase and ACC oxidase transcripts.

Appendix 4: Coded features from Discussion section in a Biology paper (930 words)

General Discussion

Although corporate associations, particularly in corporate image research, have a long history in marketing, **we found** limited empirical evidence in the literature on the relationship between corporate associations and brand-level responses. Recently, however, there have been suggestions by brand theorists that a link may exist between product judgments and organizational associations (Aaker 1996) or secondary associations, one of which is the company that produced the product (Keller 1993). Our goal was to begin to systematically explore the influence of corporate associations on consumer product evaluations. We also sought to differentiate between two distinct types of corporate associations-CA associations and CSR associations-and investigate the nature of the influence that each might have on new product evaluations.

One important finding of our research is the empirical validation of the relationship between corporate associations and consumer product responses. That is, what consumers know about a company can influence their reactions to the company's products. The implication for marketing managers is straightforward and offers confirmation for what many may already believe (ref): Paying attention to and managing all the associations that people have about a company, both for abilities and social responsibility, is an important strategic task.

Although both general types of corporate associations can be influential, in our studies, **we found** that a reputation based on a company's abilities may have a greater impact on both specific product attribute perceptions and the overall corporate evaluation than a reputation for social responsibility**. We found** that CA associations can exert duel influences on evaluations of new products through their effect on (1) product attribute perceptions and (2) the overall corporate evaluation. **We also found** that CSR associations appear to influence on the corporate evaluation. Consequently, **another important contribution of this research** is the identification and validation of multiple paths of influence for corporate associations.

In many situations, important product attributes cannot be fully evaluated prior to purchase; at the time of purchase, information is effectively missing about these attributes. The results of all three studies indicate that consumers **can and will** use CA associations as the basis for inferences about missing product attributes. Thus, through the development of CA associations, marketing managers can leverage what consumers know about a company to compensate for what they do not know and cannot evaluate about a product.

In addition, CA associations can influence new product evaluations through their effect on how consumers feel overall about the company. The corporate evaluation, or attitude toward the company, exhibited an influence on the product evaluation that was independent of the influence of CA associations on specific product attributes. Thus, even in situations in which product attribute levels are known prior to purchase and consumption, a company may still derive value from the CA associations that consumers possess.

Although the previous discussion highlights the importance of managerial attention to CA associations, our results also suggest that CSR associations have a significant influence on consumer responses to new products. **The results of all three studies demonstrate** that negative CSR associations ultimately can have a detrimental effect on overall product evaluations, whereas positive CSR associations can enhance the product evaluations.

Marketing managers have been encouraged to pursue "enlightened self-interest" by striving to achieve various societal goals while earning profits. For example, some authors suggest using cause-related marketing as an effective tool for doing societal good and enhancing company profits (ref). To date, however, there is little evidence suggesting how societally oriented activities might bring about positive outcomes for the firm. When consumers know about such activities, our research indicates that CSR associations influence the overall evaluation of the company, which in turn can affect how consumers evaluate products from the company. All else being equal, more positive evaluations should produce greater revenues for a firm Although we allowed for the possibility that CSR associations might have a direct effect on evaluations of socially related product attributes in Study One and Study Two, we did not observe such an influence. Thus, though it appears that the primary influence of CSR associations comes through their influence on the corporate evaluation rather than through any influence on specific product attributes, they still **must be** an important consideration in strategic decisions.

Finally, **Studies One and Two demonstrate** that there may be ways for managers to partially overcome the effects of negative corporate associations on product evaluations. The results indicate that when there is a discrepancy between the evaluative implications of the corporate exert an influence on product evaluations through their associations and the new product (e.g., poor CA associations and good product), a contrast effect can occur, which causes the evaluation of the product to be higher when it is produced by a company with more negatively evaluated CA associations than when it is produced by a company with more positively evaluated CA associations. One implication of our research, then, is that it may be possible for companies with a poor reputation based on CA associations to overcome (or actually benefit from) the expected detrimental effects on product evaluations by introducing truly good products. In short, the new product may be evaluated especially highly in light of its corporate context. Similarly, based on the results of Study Two, it appears that a new product introduced by a company with positive product. corporate associations may receive lower evaluations than it might otherwise have received. However because corporate associations influence product responses through multiple routes, it is still unclear under which circumstances the influence of the corporate evaluation on product evaluations (i.e., the observed contrast effect) outweighs the influence of corporate associations on product attribute inferences.

References

Bazerman, C. 1988. *Shaping written knowledge*. Madison: University of Wisconsin Press.

Becher, T. 1989. *Academic tribes and territories: intellectual inquiry and the cultures of disciplines*. Milton Keynes, SRHE/Open University Press.

Bizzell, P. 1992. *Academic discourse and critical consciousness*. Pittsburgh: University of Pittsburgh Press.

Bloor, T. 1996 Three hypothetical strategies in philosophical writing. In E. Ventola & A. Mauranen (eds). *Academic writing: intercultural and textual issues*. (pp. 19-43). Amsterdam: John Benjamins.

Brown, P. & Levinson, S. 1987. *Politeness: Some universals in language usage*. Cambridge: CUP.

Bruffee, K. 1986. Social construction: Language and the authority of knowledge. A bibliographical essay. *College English*, 48, 773-779.

Channell, J. 1990. Precise and vague quantities on writing in economics. In W. Nash (Ed.), *The Writing Scholar: Studies in Academic Discourse.* Newbury Park, Ca.: Sage.

Cohen, M. & Manion, L. 1985. Research methods in education (2nd edition). London: Croomhelm.

Dubois, B.L. 1987. "Something in the order of around forty to forty four": Imprecise numerical expressions in biomedical slide talks. *Language and Society*, 16: 527-541.

Gilbert, G. and Mulkay, M. 1984. *Opening Pandora's Box: A Sociological Analysis of Scientific Discourse*. Cambridge: CUP.

Gosden, H. (1993). Discourse functions of subject in scientific research articles. *Applied Linguistics, 14* (1), 56-75.

Halliday, M.A.K. 1978 *Language as a social semiotic: The sociological interpretation of language and meaning*.. London: Edward Arnold.

Halliday, M. (1988). On the language of physical science. In M. Ghadessey (Ed.), *Registers of Written English* (pp. 162-178). London: Pinter.

He, A. (1993). Exploring modality in institutional interactions: cases from academic counselling encounters. *Text, 13* (2), 503-528.

Holmes, J. (1983). Speaking English with the appropriate degree of conviction. In C. Brumfit (Ed.), *Learning and teaching languages for communication: Applied linguistics perspectives*. London: BAAL.

Holmes, J. 1984 Modifying illocutionary force. *Journal of Pragmatics, 8* , 345-365.

Holmes, J. 1988 Doubt and certainty in ESL textbooks. *Applied Linguistics*, 9, 20-44.

Holmes, J. (1990). Hedges and boosters in women's and men's speech. *Language and communication, 10* (3), 185-205.

Hyland, K. 1996a. Writing without conviction? Hedging in science research articles. *Applied Linguistics*, 17, 4, 433-454.

Hyland, K. 1996b. Talking to the academy: Forms of hedging in science research articles. *Written Communication*, 13, 2, 251-281.

Hyland, K. 1997. Scientific claims and community values: articulating an academic culture. *Language and Communication*, 16, 1: 19-32.

Hyland, K. 1998a. *Hedging in scientific research articles*. Amsterdam: John Benjamins.

Hyland, K. 1998b. Persuasion and context: the pragmatics of academic metadiscourse. *Journal of Pragmatics.*

Hyland, K. forthcoming. Disciplinary discourses: writer stance in research articles. In C. Candlin & K. Hyland (eds.) *Writing: texts, processes and practices*. London: Longman.

Hyland, K. & Milton, J. 1997 Hedging in L1 and L2 student writing. *Journal of Second Language Writing.* 6, 2: 183-206.

Kolb, D. A. 1981. Learning styles and disciplinary differences. In A. Chickering ed. *The modern American College*. pp 232-255. San Francisco: Jossey Bass.

Kuhn, T. 1970 *The structure of scientific revolutions. 2nd ed*. Chicago: University of Chicago Press.

Latour, B. and Woolgar, S. 1979. *Laboratory life: The social construction of scientific facts*. Beverly Hills: Sage.

Lyons, J. 1977. *Semantics. Vols 1 & 2*. Cambridge: Cambridge University Press.

MacDonald, S. 1994. *Professional academic writing in the humanities and social sciences*. Carbondale: Southern Illinois University Press.

Myers, G. 1989. The pragmatics of politeness in scientific articles. *Applied Linguistics*, 10, 1-35.

Nash, W. 1990. Introduction: The stuff these people write. In W. Nash (Ed.), *The Writing Scholar: Studies in Academic Discourse.* Newbury Park, Ca.: Sage.

Quirk, R., Greenbaum, S., Leech, G. & Svartvik, J. 1972. *A Grammar of Contemporary English*. Harlow, Essex: Longman.

Rorty, R. 1979. *Philosophy and the mirror of nature*. Princeton: Princeton University Press.

Salager-Meyer, F. 1994. Hedges and textual communicative function in medical English written discourse. *English for Specific Purposes, 13,* 2, 149-70.

Skelton, J. 1988b. Comments in academic articles. In P. Grunwell (Ed.), *Applied linguistics in society* . London: CILT/BAAL.

Skelton, J. 1997. The representation of truth in academic medical writing. *Applied Linguistics*. 18 (2) 121-140.

Swales, J. (1990). *Genre Analysis: English in academic and research settings*. Cambridge: CUP.